Venkata Lakshmi Parimala Pasupuleti

Solutions Architect · GenAI · Foundation Models · RAG · Startup Advisory · AWS · Agentic AI · Python · Technical Content · 9yr Production

757-656-9038 | venkatalakshmigmu@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Solutions Architect and GenAI practitioner with 9+ years of production engineering and hands-on experience building production-ready AI-powered products - the technical advisor early-stage founders need in their corner. I build and evaluate generative AI applications daily: foundation model selection across Anthropic, OpenAI, and Ollama with explicit cost and latency tradeoffs; RAG pipelines with vector database retrieval; multi-agent orchestration with LangGraph; and AI safety guardrails with HITL controls. AgentProbe is the open-source proof point - a multi-provider AI evaluation platform surfacing model accuracy, hallucination rate, latency, and token efficiency so builders can make data-driven model choices. LeaseI is the production multi-agent system: 8-agent LangGraph StateGraph with RAG, shared memory, tool invocation, and deterministic guardrails for regulated decisions. At Magizhs I was the trusted technical advisor for an external startup - sole owner from requirements to production deployment, making the architectural decisions that set them up to move fast without creating bottlenecks. As a Graduate Teaching Assistant I delivered technical workshops to 600+ students and translated complex AI and data engineering concepts into actionable guidance. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

GenAI & Foundation Models: Python (9yr), Generative AI, Foundation Models, LLM Selection, Model Evaluation, Anthropic API, OpenAI API, Ollama, AWS Bedrock-compatible, Multi-Provider AI Routing, Prompt Engineering, Few-Shot Prompting, Chain-of-Thought, Context Engineering, AI Safety, HITL Guardrails, Hallucination Detection, AI Best Practices, Responsible AI, Cost Optimization, Latency Optimization, Model Accuracy

AI Architecture & Solutions: Solutions Architecture, RAG Pipelines, Vector Databases (ChromaDB), Semantic Search, Embeddings, Data Pipelines, Fine-Tuning Workflows, LangChain, LangGraph, MCP, Multi-Agent Orchestration, Agentic AI, Tool Calling, Memory Management, FastAPI, REST APIs, API Design, Reference Architectures, Technical Content Creation, Developer Experience, Technical Workshops

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue, Lambda, DynamoDB), Cloud Computing, Serverless, Docker, Microservices, RabbitMQ, GitHub Actions CI/CD, PostgreSQL, MySQL, Oracle, Redis, React.js, TypeScript, Linux, Git, Agile/Scrum, Startup Advisory, Cross-functional Delivery, Stakeholder Communication

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Startup Technical Advisor (External Client)

- Embedded with early-stage startup as sole technical owner and trusted technical advisor - led architectural discovery, selected foundation models and AI services, designed multi-agent LangGraph agentic workflow, built RAG retrieval layer, implemented HITL guardrails for regulated decisions, and delivered production deployment with observability infrastructure in weeks. Made the pragmatic architectural decisions that set the customer up to move fast without creating bottlenecks.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Delivered technical workshops and developer enablement to 600+ students across Python, SQL, data engineering, and AI/ML - translating complex technical concepts into actionable guidance for diverse audiences; built ML pipelines (PyTorch, scikit-learn, R^2=0.977); designed observability dashboards improving research comprehension 40%; cross-functional technical delivery across 3 research pipelines.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Production systems at cloud scale** - Python, FastAPI/Flask REST APIs, AWS S3/EC2/RDS/Glue/Lambda, Docker microservices redesign (50% scalability improvement, 30% deployment time reduction); RabbitMQ event-driven async with retry logic and idempotency guarantees; PostgreSQL/MySQL/Oracle; owned systems for 7 years at 99.5% uptime across 100K+ users; 200+ SQL optimizations (30% performance gain); GitHub Actions CI/CD; 100% PHI compliance on AWS S3 document migration.
- **Technical leadership and developer enablement** - led projects spanning multiple engineers; wrote design docs and reference architectures; mentored 5+ engineers on system design and AWS best practices; 300+ peer code reviews; built internal tooling reducing debugging time 40%; Python/Bash automation eliminating 80% of manual workload; delivered technical guidance to product, compliance, and operations stakeholders across 120+ Agile sprints.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: GenAI Evaluation Platform - Foundation Model Selection & AI Best Practices

GenAI · Multi-Provider · Cost/Latency · RAG · Open Source

Python · FastAPI · LangGraph · ChromaDB · MCP · GitHub Actions · Anthropic API · OpenAI API · Ollama · AWS Bedrock-compatible | github.com/venkatalakshmiparimala/AgentProbe

Open-source GenAI evaluation platform designed to help AI/ML startups make data-driven foundation model decisions - exactly the tooling early-stage founders need. **Foundation model selection:** multi-provider routing (Anthropic, OpenAI, Ollama, AWS Bedrock-compatible) with explicit cost and latency tradeoffs at every inference call. **AI best practices engine:** 5-metric quality evaluation (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-Judge scoring; pass@k multi-trial harness for non-deterministic outputs. **RAG & data pipelines:** ChromaDB vector database retrieval, semantic search, offline batch evaluation pipelines. **Platform:** 15-endpoint FastAPI backend; CI regression gate (GitHub Actions); reference architecture for production-ready AI evaluation. Open-source. Zero regressions across 1,240+ test cases.

Leasel: Production-Ready GenAI Application - Multi-Agent + RAG + Safety

Production AI · Multi-Agent · RAG · Vector DB · HITL Safety · LangGraph

Python · LangGraph (8-agent) · FastAPI · RAG · ChromaDB · SQLite · MCP · Anthropic API | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production-ready GenAI application demonstrating the architecture patterns AWS startup founders need to build and scale AI-powered products. **Foundation model integration:** Anthropic API with structured prompting, confidence scoring, and output validation - managing accuracy and safety for regulated decisions. **RAG & vector database:** ChromaDB semantic retrieval over regulatory and operational knowledge corpora; citation-backed outputs with groundedness scoring. **Multi-agent orchestration:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation) with conditional routing based on confidence scores. **AI safety:** deterministic HITL escalation blocks automated resolution of high-risk decisions - the guardrails early-stage AI products require. 93.6% precision, 87.9% F1 across 1,240 test cases.

AI on FHIR: Production GenAI Application - Regulated Healthcare Data

GenAI App · AWS · HIPAA · API Integration · Startup Architecture

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · BERT · spaCy · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Production-ready GenAI application on AWS for a regulated healthcare environment - the reference architecture pattern for AI/ML and HealthTech startups building on AWS. LLM-powered tool invocation translates natural-language queries into structured FHIR API calls; HIPAA-compliant RBAC with full data isolation and audit trail; AWS Lambda serverless backend with API gateway; ~60% reduction in manual retrieval time. Demonstrates the pragmatic startup architecture: move fast, maintain compliance, build to scale.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017